

JACK NAIMER

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EDUCATION & TRAINING

Engineering Science – Machine Intelligence Major, University of Toronto 2021-2025
▪ Dean's Honours List – Winter 2024, Fall 2023, Fall 2021

Certificate, Math and Computer Science (MaCS) enriched high school program, William Lyon Mackenzie Collegiate Institute 2017-2021

High School Diploma, William Lyon Mackenzie Collegiate Institute 2021

WORK EXPERIENCE

Undergraduate Researcher, Bo Wang Lab, University Health Network. March 2024-Present
Writing undergraduate thesis on generative AI DNA language models.
Developing and deploying a machine learning model to predict surgical durations and improve operating room scheduling.

Undergraduate Researcher, Angela Schoellig Learning Systems and Robotics Lab (LSY Lab), Technical University of Munich, Germany. May-August 2024
[Paper Preprint](#) - Developed novel AI-based robotic semantic safety filter.
Developed a Lego-building robot using Franka robot arm. Focused on AI-vision components including brick detection and pose estimation.

Data Science Intern, Marsh New York Office. June-August 2023
Developing machine learning and statistical models to improve risk analysis and forecasts for Marsh clients. Focused on risk analysis in the healthcare space, developing new machine learning models to assess risks of hospitals.

Research Student, SickKids Hospital with Dr. Sasha Litwin. May-June 2023
Developing an AI chatbot designed to answer a parent's fever questions, improve at-home treatment, and prevent unnecessary ER visits.

Research Student, Dr. Mike Fralick's Lab, Mount Sinai Hospital. May 2022-May 2023
Developing AI tools for healthcare.

Projects

ChatBotMD	Medical Literature Tool	jrnowl.com
- Created BERT-based AI chat bot prototype for clinical use - Developed custom decision-making algorithms - Created chat bot website from scratch, set up networking, and run on AWS server with NGINX - Sole programmer	- AI tool to inform doctors of advances in medicine - Finds relevant medical publications and gives concise summaries - Created react-js website - Sole programmer	Using ML to automatically find journal publishing guidelines from medical journal websites for use on jrnowl.com

Bike Service Technician, Skiis & Biikes. Serviced and repaired bicycles, trouble-shoot issues, communicate with customers. May-Sept 2021

Sales, Gears Bike Shop Leaside. Direct customer service, worked the cash. June-Oct 2020

Hockey Referee, Forest Hill Hockey Association. 2017-2020

Camp Counsellor, Green Acres Day Camp. Led activities for campers age 5-8 Summer 2019

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EXTRACURRICULAR & VOLUNTEER EXPERIENCE

Academic Director & Academic Paper Read Group Mentor, University of Toronto Machine Intelligence Student Team (UTMIST). 2022-2023

Lead monthly lectures explaining complex machine learning papers and topics in an easy-to-understand way. Topics taught include CNN, ConvNext, GNN, Variational Autoencoder, Attention, Transformer, etc.

Software Team Member, University of Toronto Design League Robodog Team. Responsible for inverse kinematics, and gait programming using ROS. 2022

Competition Team Member, University of Toronto Concrete Toboggan Design Team. 2021-2022

Satirical Newspaper Co-President, William Lyon Mackenzie CI. 2020-2021

Math and Computer Science (MaCS) Program Peer Mentor and Math Peer Tutor, William Lyon Mackenzie Collegiate Institute. 2019-2021

TECHNICAL SKILLS

- Python, Julia, C, C++, Java, Matlab, Linux, ROS, Docker, HTML, CSS, SolidWorks
- Deep Learning, Computer Vision, LLMs, PyTorch, JAX, Tensorflow, NLP
- University of Toronto Machine Shop Certification (2021)

AWARDS & ACHIEVEMENTS

Engineering Science Global Research Opportunities Program, For summer research in Angela Schoellig's LSY Lab. 2024

BioTalent Health and Biosciences Student Work Placement Wage Subsidy, For summer research in Dr. Mike Fralick's Lab. 2022

Admission Scholarship, Faculty of Applied Science and Engineering, University of Toronto. \$2000 2021

Advanced Placement (AP), 5/5 Computer Science A (2019), 5/5 Calculus AB(2021)

DECA Provincial Finalist & Case Award, Business Finance Series 2021

Honourable Mention, Queen's Model United Nations Conference 2021

3rd Place, LyonHacks Hackathon, William Lyon Mackenzie Collegiate Institute 2021

DECA Regional Top 60, Business Finance Series 2020

HOBBIES

Mountain Biking Daily rides in Toronto's Don Valley trails year-round.

Guitar Weekly lessons from age 6-18. Practice frequently.

Martial Arts Brazilian Jiu Jitsu weekly lessons at The University of Toronto. Taekwondo Black Belt and dedicated to personal practice.

Swing Dancing Participate in weekly lessons and social swing dances to live jazz music.

Ultimate Frisbee University of Toronto intramurals. Played daily in high school.

Hockey Participate in local pick-up games several times per week. Goalie and referee at Forest Hill Hockey Association until 2020.